Kinematics (Extended)

ANSWER KEY

Core Concepts: Kinematics (Extended)

1 For the model $f(x) = 3x + 4$ in kinematics (extended), evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

3 A data set recorded for kinematics (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 3x + 4$ in kinematics (extended), evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

6 A data set recorded for kinematics (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

Graphical & Geometric Interpretation

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7 For the model $f(x) = 3x + 4$ in kinematics (extended), evaluate $f(4)$.

a. 16 ✓

b. 19

c. 12

d. 20

Substitute $x = 4$: $f(4) = 3(4) + 4 = 16$.

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

9 A data set recorded for kinematics (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 3x + 4$ in kinematics (extended), evaluate $f(5)$.

a. 19 ✓

b. 22

c. 15

d. 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ **and** $x = 9$ ✓

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

12 A data set recorded for kinematics (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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